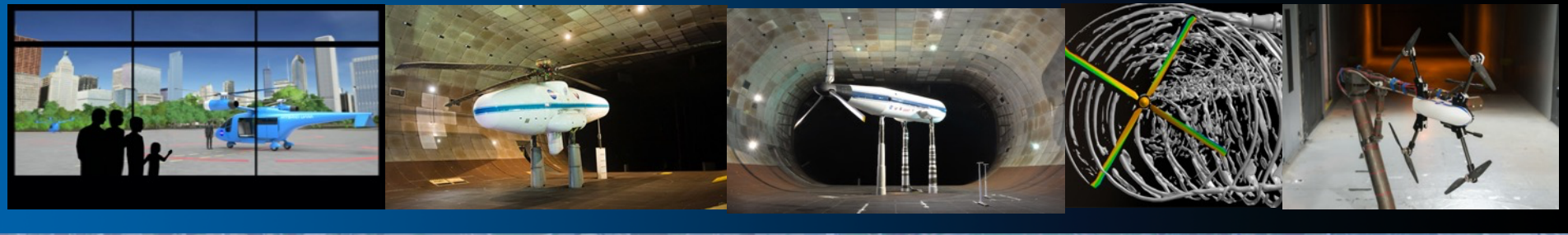
Status of NASA/Army Hover Test

Thomas R. Norman

NASA Ames Research Center

AIAA APATC Rotorcraft Discussion Group Meeting

January 13, 2021



Outline

- Covid Effects on Test Schedule
- CY20 Accomplishments
- Next Steps

- Backup Slides
  - Test Description and Key Measurements
  - Data Products for CFD Validation

Covid Effects on Test Schedule

- NASA/Army not allowed to work onsite at either LaRC or ARC between March and September 2020
  - Limited to monitoring contracted efforts and working on test plans/analyses
  - Required hardware upgrades and system checkouts delayed significantly
- Test start in 80x120 test section delayed from 9/1/20 to 3/23/21
  - Test completion expected in late June 2021
- Estimated data availability for correlation will depend on data type
  - Performance and pressure data will be available first– August 2021
  - Transition, rotor wake, blade deflection data will follow depending on data reduction progress – September to November 2021

CY20 Accomplishments

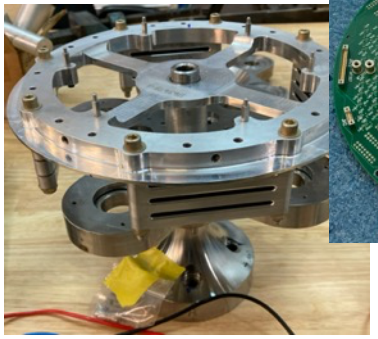
- Army Rotor Test Stand (ARTS)
  - Test stand successfully demonstrated to full power during forward-flight test in LaRC 14x22 wind tunnel
  - Based on 14x22 test results, modifications to ARTS control system designed, fabricated, and installed
  - Instrumentation hat and interface to Rotating Data Acquisition System (RDAS) fabricated and delivered



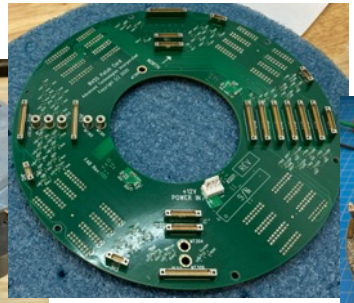
ARTS in LaRC 14x22



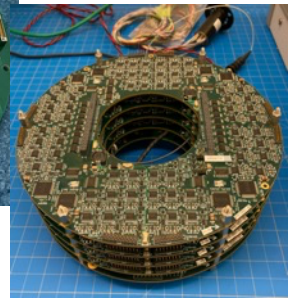
New Control System



RDAS Mount Base



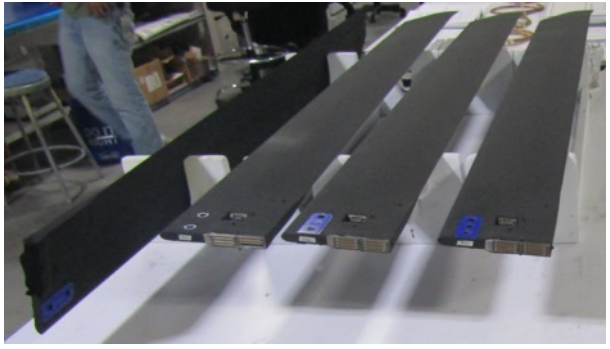
RDAS Patch Card



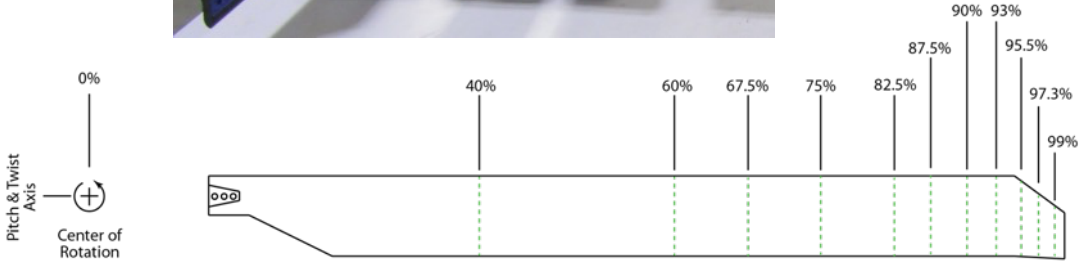
RDAS

CY20 Accomplishments

- HVAB Blades
  - HVAB blades (including pressure blade) fabricated, calibrated, and delivered
  - 172 (92%) pressure transducers and 17 (81%) strain gages still functioning after fabrication
  - Blade geometry and structural properties are fully open and were published in NASA TM 2020-5002153



HVAB Blades



Blade Number	Blade Type H- Hover A- Acoustic	Blade Instrumentation								
		Strain Gages				Pressure Transducers			Tip LED	RTDs
		Beam	Chord	Torsion	Radial Stations	Total	Radial Stations	None		
SN000	Process - H							X		
SN001	Production - H	X	X	X	1			X	X	X
SN002	Production - H	X	X	X	1			X	X	X
SN003	Production - H	X	X	X	1			X	X	X
SN004	Hover - H	X	X	X	1	187	11		X	X
SN005	Strain Gage - H	X	X	X	4			X	X	X

Next Steps

- Complete full-system rotating checkout at LaRC
  - Non-rotating RDAS checkout with HVAB blades
  - Rotating RDAS C/O without blades
  - Rotating HVAB blades
  - Disassembly and ship to ARC
- Complete optical system setup and evaluations in 80x120
  - Thermography, shadowgraphy, photogrammetry, PIV
- Complete all NFAC test preparations
  - Instrumentation, software, safety, etc

- Begin testing!!!!

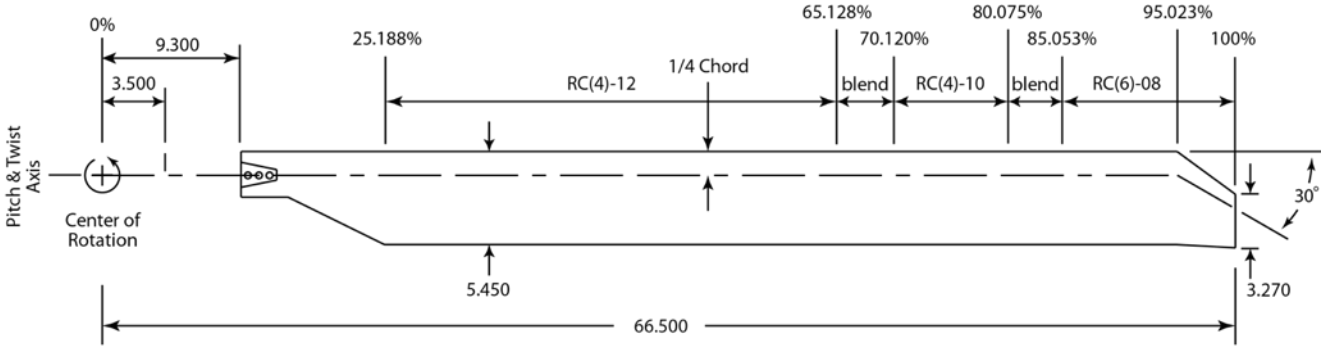


Backup Slides

- Test Description
  - Rotor System, Test Stand, Facility
  - Key Measurements
- Data Products for CFD Validation

Test Description – HVAB Blades

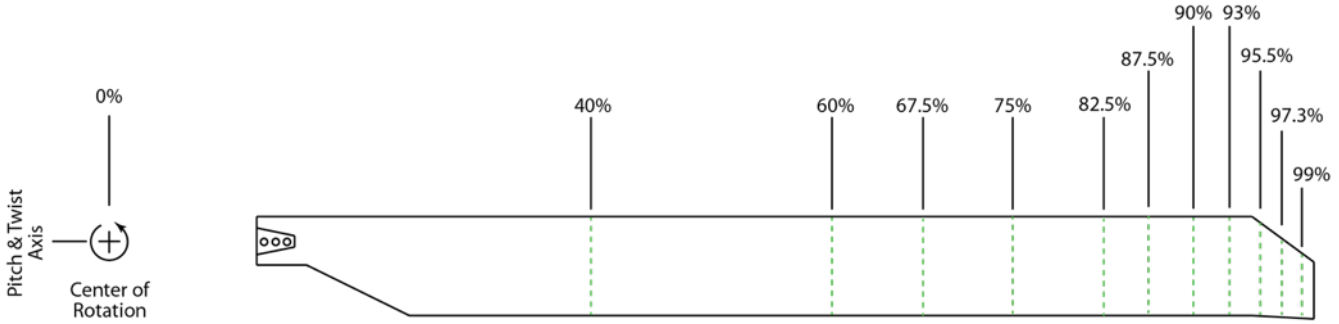
- Identical to Army PSP rotor except for root attachment location



Radius	**66.5 inches**
Reference chord	**5.45 inches**
Tip taper ratio	**60%**
Tip sweep (quarter chord)	**30 deg. at 95% R**
Airfoils	**RC series**
Number of blades	**4**

Test Description – HVAB Blades

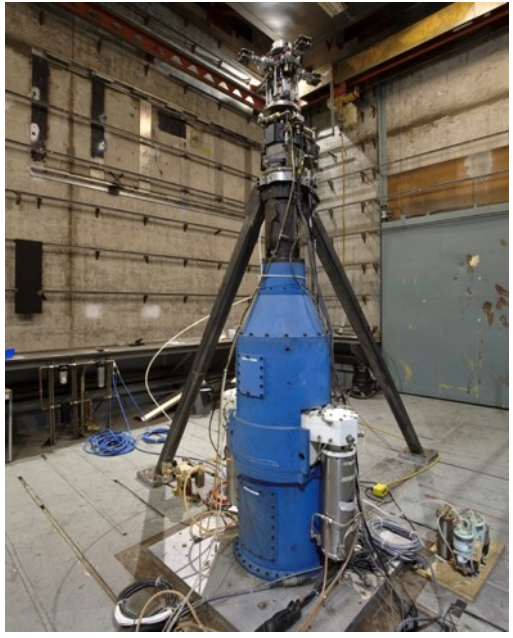
- Strain-gaged and pressure-instrumented blades
- Pressure taps (for airloads) located at 11 radial stations



Blade Number	Blade Type	Blade Instrumentation									
		Strain Gages				Pressure Transducers			Tip LED	Active Heater	RTDs
		Beam	Chord	Torsion	Radial Stations	Total	Radial Stations	None			
SN000	Process							X		X	
SN001	Production	X	X	X	1			X	X	X	X
SN002	Production	X	X	X	1			X	X	X	X
SN003	Production	X	X	X	1			X	X	X	X
SN004	Production	X	X	X	1	187	11		X		X
SN005	Production	X	X	X	1	51	14		X		X
SN006	Optional	X	X	X	4			X	X	X	X

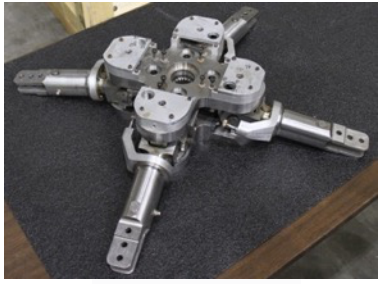
Test Description – Test Stand/Facility

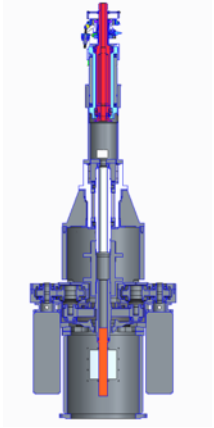
- Army ARTS test stand in NFAC 80- by 120-Foot test



ARTS Test Stand in Hover Chamber

Articulated Hub





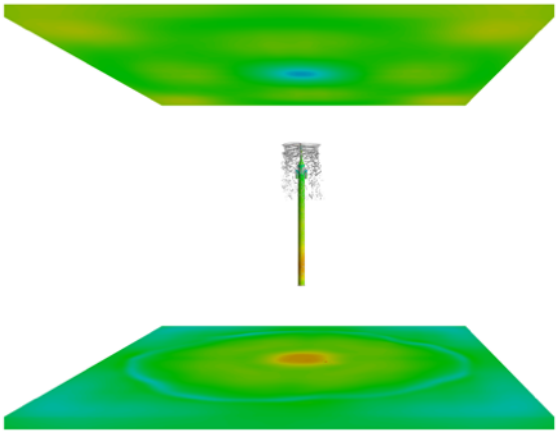
NFAC 80- by 120-Ft Test Section



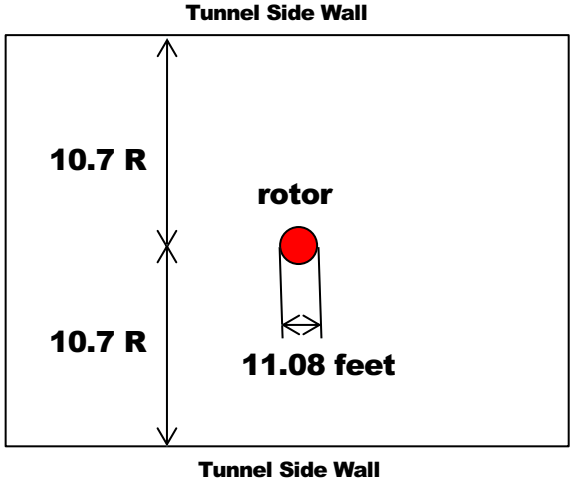
Test Description – Test Stand/Facility

- Test stand and rotor – relative scale

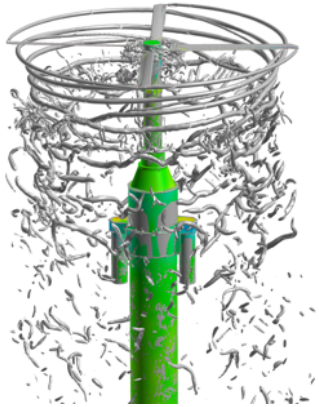
Rotor/Stand relative to floor/ceiling



Rotor relative to side walls– top view



Rotor relative to test stand



Test Description – Key Measurements

- Rotor Thrust and Torque (rotor balance and flex coupling)
- Blade Root Motion (root collective, flap, lag) and Blade Deflections (Hall effect transducers, photogrammetry)
- Blade Pressures (Kulites)
- Boundary Layer Transition Measurements (IR Thermography)
- Rotor Wake Geometry (shadowgraphy)
- Rotor Vortex Properties (particle image velocimetry)

Data Products for CFD Validation

- Initial list of data products for CFD validation has been prepared

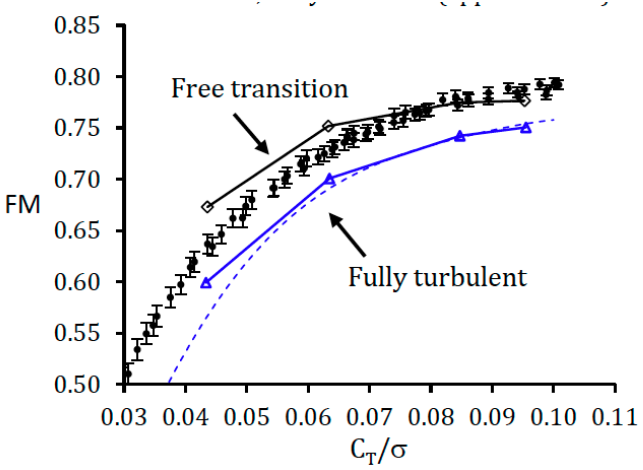
- Includes data needed to support CFD inputs as well as for validation

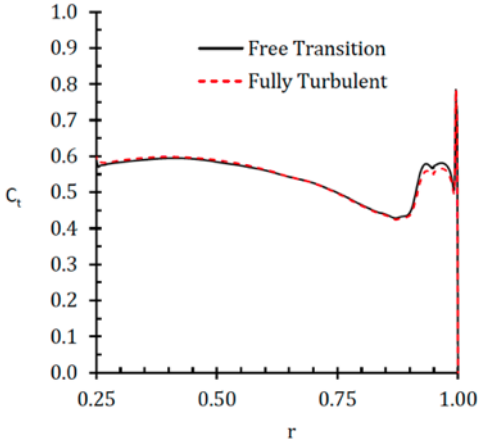
Data Products for CFD Validation

- Data for CFD Inputs
  - Test Hardware Geometry
    - CAD model/grid of rotor blade OML (as-designed and as-measured)
    - CAD model/grid of test stand and attachment hardware
    - Description of rotor hub and blade interface (final form TBD)
    - Description of 80- by 120-Foot Wind Tunnel and model location (final form TBD)
  - Rotor Operating Conditions and Blade Displacements (measured at each condition)
    - Tip Mach number (RPM, atmospheric conditions)
    - Blade root angles (blade root pitch/flap/lag)
    - Blade deflections (flap and twist vs r/R, photogrammetry) for multiple blades

Data Products for CFD Validation

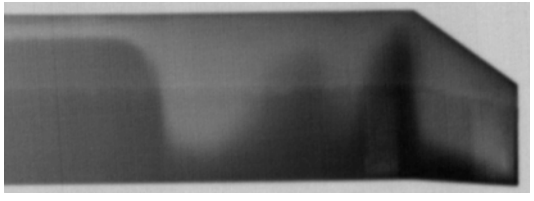
- Data for CFD Validation (at each condition)
  - Rotor Performance
    - Figure of Merit (thrust/torque, rotor balance/flex coupling) vs Ct/s
    - Ct/s vs collective
  - Blade Airloads
    - Kulite pressure transducers at 11 r/R stations, pressures plus integrated airloads

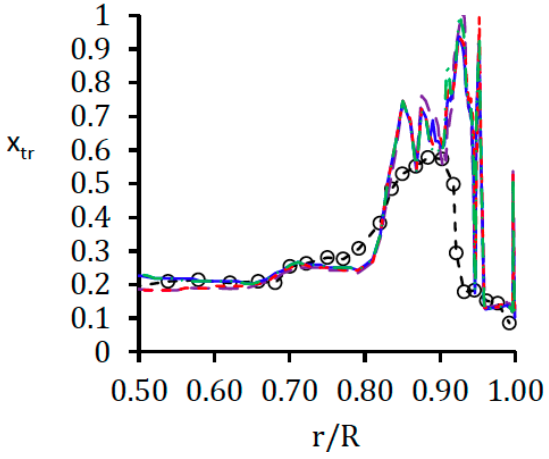


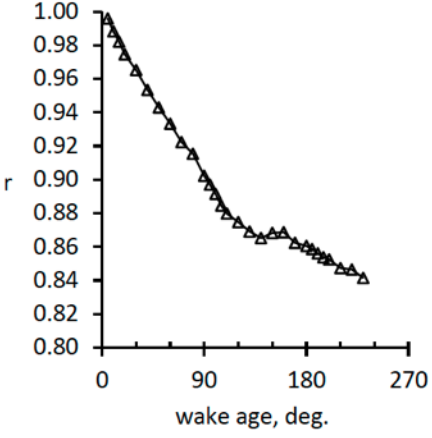


Data Products for CFD Validation

- Data for CFD Validation (at each condition)
  - Transition Location (x/c vs r/R) upper and lower surfaces (multiple blades) (IR Thermography)
  - Tip vortex geometry (z/R and r/R vs wake age) (shadowgraph)







Data Products for CFD Validation

- Data for CFD Validation (at each condition)
  - Tip vortex circulation, vorticity, and core size vs wake age (PIV)
  - Wake velocities and vorticity for different wake ages (PIV)

